

## 1. Project Overview

### 1.1 Objective

**SmartHire AI** is an AI-powered mock interview and candidate assessment platform designed to bridge the gap between candidates preparing for job interviews and recruiters seeking top talent. The platform provides tailored solutions across three primary user roles:

- **For Candidates:** An intelligent interview simulator with real-time voice recognition, facial analysis, and multi-dimensional AI scoring, helping them practice and improve their interview skills.
- **For Recruiters:** A comprehensive talent assessment dashboard with AI-ranked candidates, skill analytics, and comparison tools to make data-driven hiring decisions.
- **For Admins:** A complete governance system to manage users, monitor AI performance, and analyze platform usage.

### 1.2 Problem Statement

Traditional mock interview platforms lack essential features for effective evaluation:

1. Real-time behavioral analysis (confidence, eye contact, engagement).
2. AI-driven multi-dimensional scoring beyond simple keyword matching.
3. Comprehensive dashboards for all stakeholder roles.
4. Recruiter-candidate connection in a single unified platform.

SmartHire AI solves these issues by integrating machine learning models for facial analysis, speech recognition, and comprehensive scoring across four critical interview dimensions.

### 1.3 Key Features

| Feature                                                                                                              | Description                                                                              |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
|  <b>Voice-based Interviews</b>    | Real-time speech recognition with Web Speech API                                         |
|  <b>Live Video Recording</b>      | Webcam capture with <code>.webm</code> storage                                           |
|  <b>ML Face Analysis</b>          | Real-time confidence detection via <code>face-api.js</code> + custom Random Forest model |
|  <b>Multi-Dimensional Scoring</b> | Communication, Confidence, Technical, Professionalism                                    |
|  <b>Performance Analytics</b>     | Trend charts, skill breakdowns, weak-area predictions                                    |
|  <b>Recruiter Tools</b>           | Candidate comparison, rankings, shortlisting                                             |
|  <b>Admin Dashboard</b>           | User management, system health, AI monitoring                                            |
|  <b>Email Notifications</b>       | Interview reminders, performance reports                                                 |

| Feature                                                                                           | Description                          |
|---------------------------------------------------------------------------------------------------|--------------------------------------|
|  Report Downloads | JSON/CSV export of interview results |

## 2. Technology Stack

### 2.1 Frontend

| Technology        | Purpose                                   |
|-------------------|-------------------------------------------|
| React.js 18       | UI framework                              |
| Bootstrap 5       | Responsive styling                        |
| Chart.js          | Analytics visualizations                  |
| face-api.js       | Facial landmark detection (TensorFlow.js) |
| React Speech Rec. | Voice-to-text conversion                  |
| Web Speech API    | Text-to-speech for questions              |
| Axios             | HTTP client                               |

### 2.2 Backend

| Technology    | Purpose                         |
|---------------|---------------------------------|
| Node.js 20    | Runtime environment             |
| Express.js    | REST API framework              |
| PostgreSQL 14 | Relational database             |
| JWT           | Authentication token management |
| Passport.js   | Google OAuth 2.0 integration    |
| Bcrypt        | Password hashing                |
| Nodemailer    | Email notifications             |
| Multer        | Video file uploads              |

### 2.3 Machine Learning

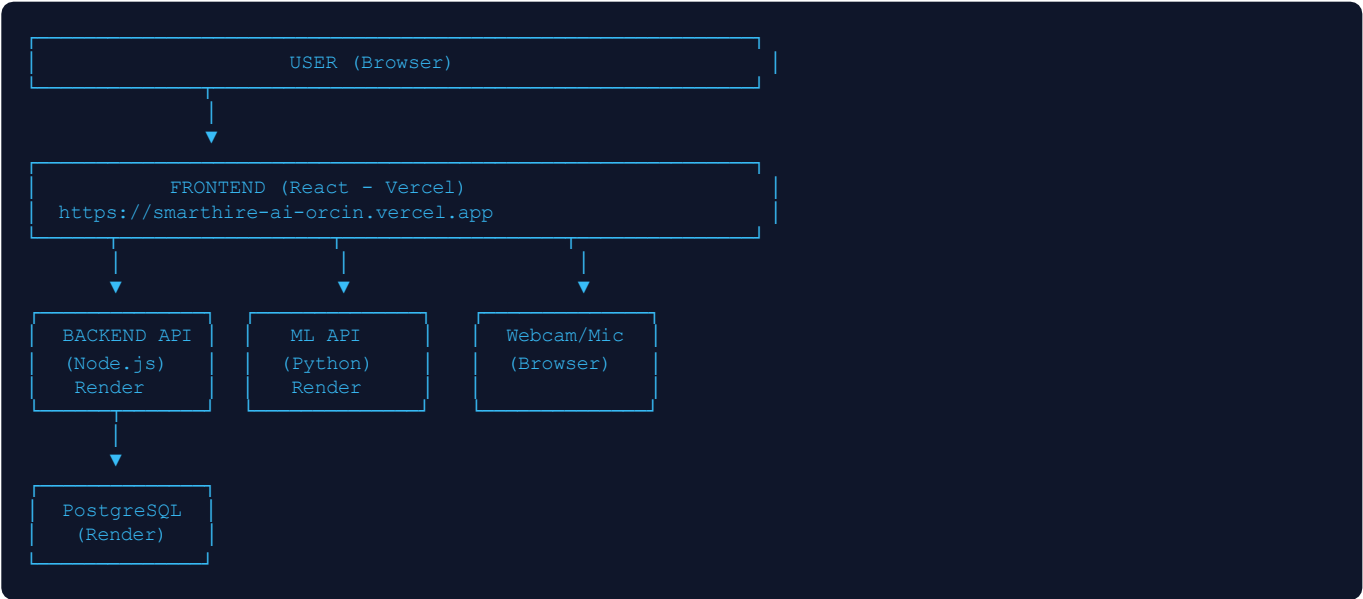
| Technology     | Purpose                  |
|----------------|--------------------------|
| Python 3.11    | ML runtime               |
| Flask          | ML API framework         |
| scikit-learn   | Random Forest classifier |
| joblib         | Model serialization      |
| NumPy / Pandas | Data processing          |

### 2.4 DevOps & Deployment

| Service      | Purpose                           |
|--------------|-----------------------------------|
| Vercel       | Frontend hosting                  |
| Render       | Backend + ML + PostgreSQL hosting |
| GitHub       | Version control                   |
| Google Cloud | OAuth credentials                 |

### 3. System Architecture & Data Flow

#### 3.1 High-Level Architecture



#### 3.2 Data Flow

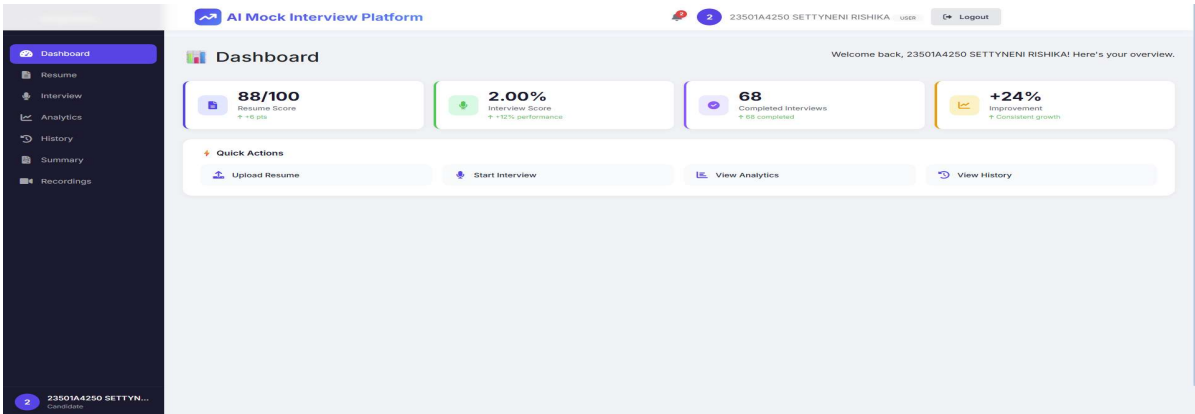
- Authentication:** User logs in → Backend issues JWT → Frontend stores token securely.
- Interview Generation:** Interview generated → Backend fetches questions from question bank based on round & domain.
- Session Capture:** Interview session active → Frontend captures voice + video live streams.
- Live ML Stream:** Every 2 seconds, facial feature landmarks are processed and sent to the Python ML API.
- Submission & Scoring:** Answers + behavioral metadata sent to backend → Calls scoring engine → Stores structured feedback in DB.
- Analytics & Reports:** Recruiter/Admin views populated analytics in multi-role dashboards.

### 4. Candidate Dashboard

Empowers job seekers to practice interviews, review real-time feedback, and track improvement over time.

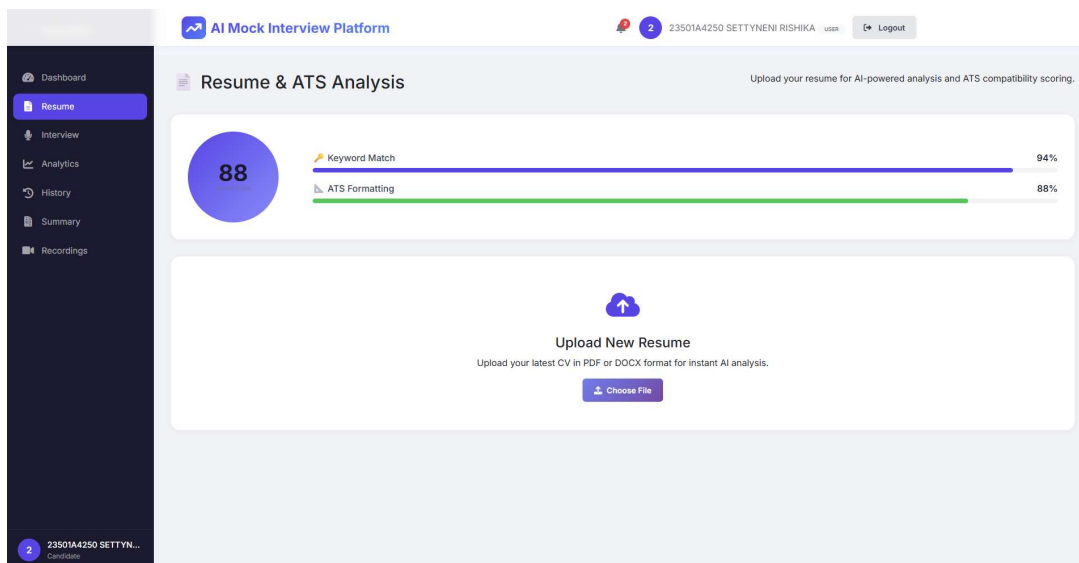
#### 4.1 Dashboard Overview

The main overview displays Resume Score (0-100), Latest Interview Score, Completed Interviews Count, and Improvement Trend percentages, along with quick actions for setup.



## 4.2 Resume Tab :

Provides AI-powered resume scoring, ATS compatibility checks, keyword matching analysis, and structural formatting scoring.

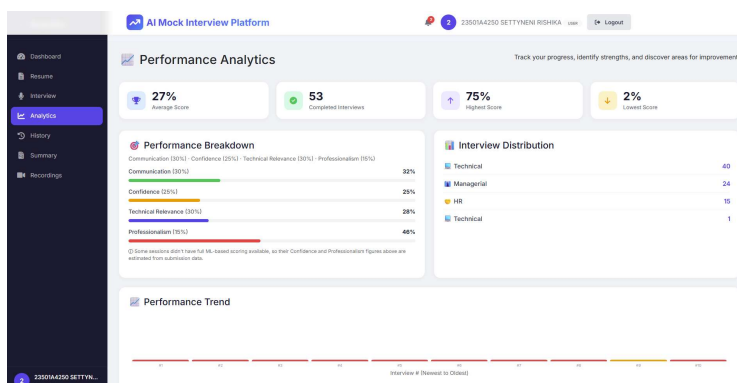


## 4.3 Interview Tab (Core Feature)

- **New Interview Setup:** Choose round (TR/MR/HR), domain, difficulty, and question count.
- **Live Voice Interview:** AI reads questions aloud (Web Speech API), candidate answers via microphone with real-time transcription.
- **Live ML Widget (Floating bottom-right):** Displays confidence level (Composed / Steady / Needs Focus), eye contact detection, attention/engagement scores, and head-turn violation warnings (auto-terminates after 3 violations).
- **Speech Analysis:** Pace (WPM), filler word count, grammar, and pronunciation scores.

## 4.3 Analytics, History, Summary & Recordings Tabs

- **Analytics Tab:** Performance stats (Avg/Highest/Lowest), skill radar breakdown, distribution charts, and strengths/weaknesses recommendations.
- **History Tab:** Filterable table of past interviews with options to view detailed report, download JSON, or email report.
- **Summary & Recordings:** Long-term trend filters (30D, 90D, All), skill progress bars, and video recording playback/management.



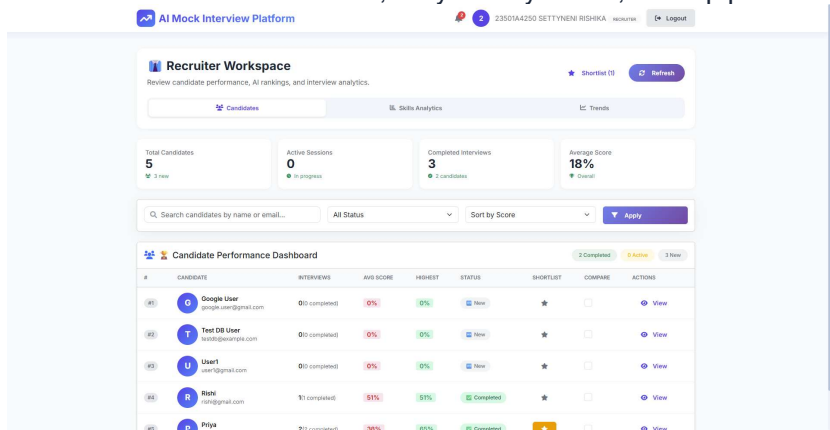
The screenshot shows the 'Interview History' dashboard. It features a sidebar with navigation links: Dashboard, Resume, Interview, Analytics, History (active), Summary, and Recordings. The main content area displays a table of past interviews with columns: Round, Domain, Difficulty, Status, Score, Duration, Date, and Action. The table lists several interviews with details such as 'Technical', 'side', 'Easy', 'Completed', '2.00%', '1m 15s', and 'Sep 10, 2026, 03:27 PM'.

| ROUND | DOMAIN     | DIFFICULTY         | STATUS | SCORE     | DURATION | DATE    | ACTION                 |
|-------|------------|--------------------|--------|-----------|----------|---------|------------------------|
| 1     | Technical  | side               | Easy   | Completed | 2.00%    | 1m 15s  | Sep 10, 2026, 03:27 PM |
| 2     | Managerial | Leadership         | Easy   | Completed | 60.00%   | 1m 21s  | Sep 4, 2026, 07:04 PM  |
| 3     | Managerial | project_management | Easy   | Completed | 0.00%    | 1m 15s  | Sep 4, 2026, 06:02 PM  |
| 4     | Managerial | project_management | Easy   | Completed | 35.00%   | 0m 56s  | Sep 4, 2026, 05:59 PM  |
| 5     | Technical  | side               | Easy   | Completed | 7.00%    | 0m 28s  | Sep 4, 2026, 04:44 PM  |
| 6     | Technical  | general            | Easy   | Completed | 3.00%    | 0m 43s  | Sep 3, 2026, 03:50 PM  |
| 7     | Technical  | side               | Easy   | Completed | 12.00%   | 0m 33s  | Sep 3, 2026, 08:56 PM  |
| 8     | Technical  | side               | Easy   | Completed | 42.00%   | 38m 24s | Sep 3, 2026, 08:16 PM  |
| 9     | Managerial | Team_management    | Easy   | Completed | 10.00%   | 0m 38s  | Sep 3, 2026, 07:14 PM  |
| 10    | Technical  | side               | Easy   | Completed | 49.00%   | 2m 1s   | Sep 2, 2026, 08:48 PM  |
| 11    | Technical  | side               | Easy   | Completed | 16.00%   | 2m 15s  | Aug 31, 2026, 07:50 PM |
| 12    | Technical  | side               | Easy   | Completed | 0.00%    | 0m 48s  | Aug 31, 2026, 07:40 PM |
| 13    | Technical  | side               | Easy   | Pending   | -        | N/A     | Aug 31, 2026, 07:37 PM |

## 5. Recruiter & Admin Dashboards

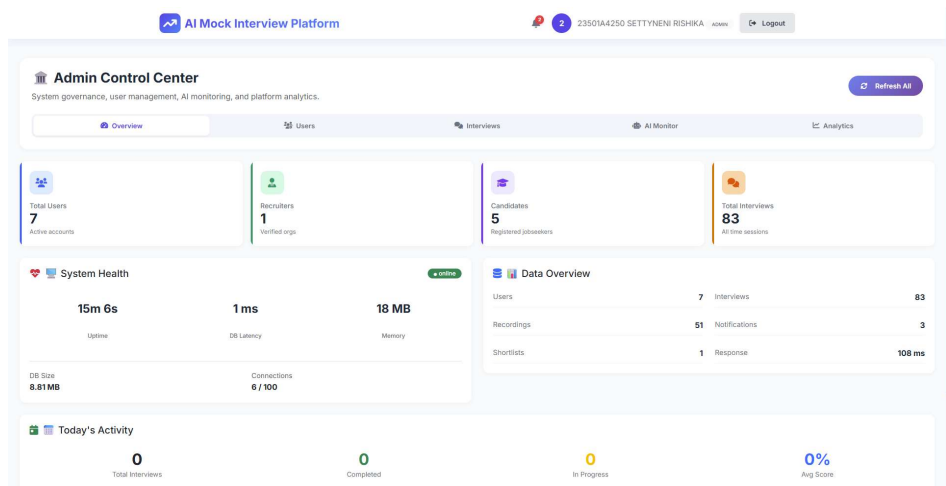
### 5.1 Recruiter Dashboard

- **Candidates Tab:** Summary statistics, candidate table sorted by score/status, shortlist management, and full candidate profile modals.
- **Candidate Comparison:** Side-by-side comparative analysis of 2-4 candidates across 4 dimensions with automated winner badges.
- **Skills Analytics & Trends:** Domain breakdowns, daily activity charts, and top performer rankings.



### 5.2 Admin Dashboard

- **Overview & System Health:** Server status, uptime metrics, DB latency, active connections, and recent activity logs.
- **User Governance & Interview Monitoring:** Role management (Candidate/Recruiter/Admin), user deletion, and real-time live interview session tracking.
- **AI Monitor Tab:** ML API health, score distribution histograms (Excellent/Good/Average/Needs Improvement), coverage %, and score variance statistics.



## 6. Machine Learning Integration

### 6.1 Confidence Detection Model

- **Dataset:** Confidence-Posture-Dataset
- **Algorithm:** Random Forest Classifier (~85% accuracy)
- **Features Extracted (15 Landmark Metrics):** Eye-shoulder ratio, shoulder span, spine angle, head tilt angle, eye distance ratio.
- **Output Classes:** Confident, Composed, Steady, Nervous, Scared, Confused.

### 6.2 Multi-Dimensional Scoring Engine

Evaluations are calculated across 4 weighted dimensions:

| Dimension       | Weight | Evaluation Factors                                                      |
|-----------------|--------|-------------------------------------------------------------------------|
| Communication   | 30%    | Clarity, grammar, filler word ratio, pace (WPM), completeness           |
| Confidence      | 25%    | Eye contact, posture stability, hesitation frequency, speech confidence |
| Technical       | 30%    | Accuracy, keyword relevance, problem solving, domain depth              |
| Professionalism | 15%    | Time management, response organization, etiquette                       |

$$\text{Overall Score} = (\text{Communication} \times 0.30) + (\text{Confidence} \times 0.25) + (\text{Technical} \times 0.30) + (\text{Professionalism} \times 0.15)$$

## 7. Notifications, Security & Deployment

### 7.1 Security Implementation

- **Password Hashing:** Bcrypt with 10 salt rounds.
- **Authentication:** JSON Web Tokens (JWT) with 7-day expiry & Passport.js Google OAuth 2.0.
- **Database & Transport:** Parameterized SQL queries preventing SQL injection, strict CORS whitelisting, and secret management via environment variables.

### 7.2 Deployment Architecture

| Component   | Platform          | URL / Connection                                                                                                              |
|-------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Frontend    | Vercel            | <a href="https://smarthire-ai-orcin.vercel.app">https://smarthire-ai-orcin.vercel.app</a>                                     |
| Backend API | Render            | <a href="https://smarthireai-mockinterview-platform.onrender.com">https://smarthireai-mockinterview-platform.onrender.com</a> |
| ML API      | Render            | <a href="https://mock-interview-ml.onrender.com">https://mock-interview-ml.onrender.com</a>                                   |
| Database    | Render PostgreSQL | Internal SSL Connection                                                                                                       |

## 8. Appendix: API Endpoints Summary

| Endpoint                   | Method | Purpose                                        |
|----------------------------|--------|------------------------------------------------|
| /api/auth/register         | POST   | User registration                              |
| /api/auth/login            | POST   | User authentication & token generation         |
| /api/interviews/generate   | POST   | Generate custom interview questions            |
| /api/interviews/:id/start  | POST   | Start live session timer and ML stream         |
| /api/interviews/submit/:id | POST   | Submit complete answer payload                 |
| /api/scoring/session       | POST   | Trigger AI scoring engine calculation          |
| /api/predict/confidence    | POST   | ML API prediction endpoint for facial features |
| /api/recruiter/candidates  | GET    | Retrieve ranked candidate list for recruiters  |
| /api/admin/users           | GET    | List system users and permission levels        |